

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
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Experiment No.	PBL Part 2 – Task 5
Title	JavaScript Functions and Classes – Product Objects

1. TITLE

JavaScript Functions and Classes – Product Objects

2. AIM

To differentiate between JavaScript functions and classes and create multiple product objects with common properties and methods using a JavaScript class.

3. OBJECTIVE

- To understand JavaScript functions and classes.
 - To create objects using a class.
 - To define common properties and methods for multiple objects.
 - To use DOM manipulation to display object data.
 - To demonstrate object-oriented programming concepts in JavaScript.
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4. THEORY

A JavaScript function is a reusable block of code that performs a specific task. A class is a blueprint for creating objects with common properties and methods. The constructor() method initializes object properties, while class methods define behavior shared by all objects created from the class. In this task, a Product class is used to create multiple product objects with the common properties name and price and a common method getDetails().

5. ALGORITHM / PROCEDURE

1. Create an HTML page with a section to display product information.
 2. Define a JavaScript Product class with name and price properties.
 3. Define a getDetails() method inside the class.
 4. Create multiple Product objects using the class.
 5. Store the objects in an array.
 6. Use a loop to generate the product details dynamically.
 7. Display the generated information in the webpage.
 8. Open the page in a browser and verify the output.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>JavaScript Classes - Products</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
```

```
<h1>Product List</h1>
<div id="productList"></div>
</div>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
}

.container {
  width: 500px;
  padding: 25px;
  background-color: white;
  border: 1px solid #ccc;
  border-radius: 8px;
}

h1 {
  text-align: center;
}

.product {
  margin-bottom: 12px;
  padding: 12px;
  border: 1px solid #ddd;
  border-radius: 5px;
}
```

script.js

```
class Product {
  constructor(name, price) {
    this.name = name;
    this.price = price;
  }

  getDetails() {
    return `${this.name} - 📌${this.price.toFixed(2)}`;
  }
}

const products = [
  new Product("Pen", 20),
  new Product("Notebook", 50),
  new Product("Book", 125.50)
];

const productList = document.getElementById("productList");

products.forEach((product) => {
  const item = document.createElement("div");
  item.className = "product";
  item.textContent = product.getDetails();
  productList.appendChild(item);
});
```

7. CODE EXPLANATION

Code	Explanation
class Product	Defines a blueprint for creating product objects.
constructor(name, price)	Initializes the common properties of every product object.
getDetails()	Defines a common method that returns formatted product information.
new Product(...)	Creates separate objects using the Product class.
forEach()	Processes every product object in the array.
createElement()	Creates an HTML element dynamically.

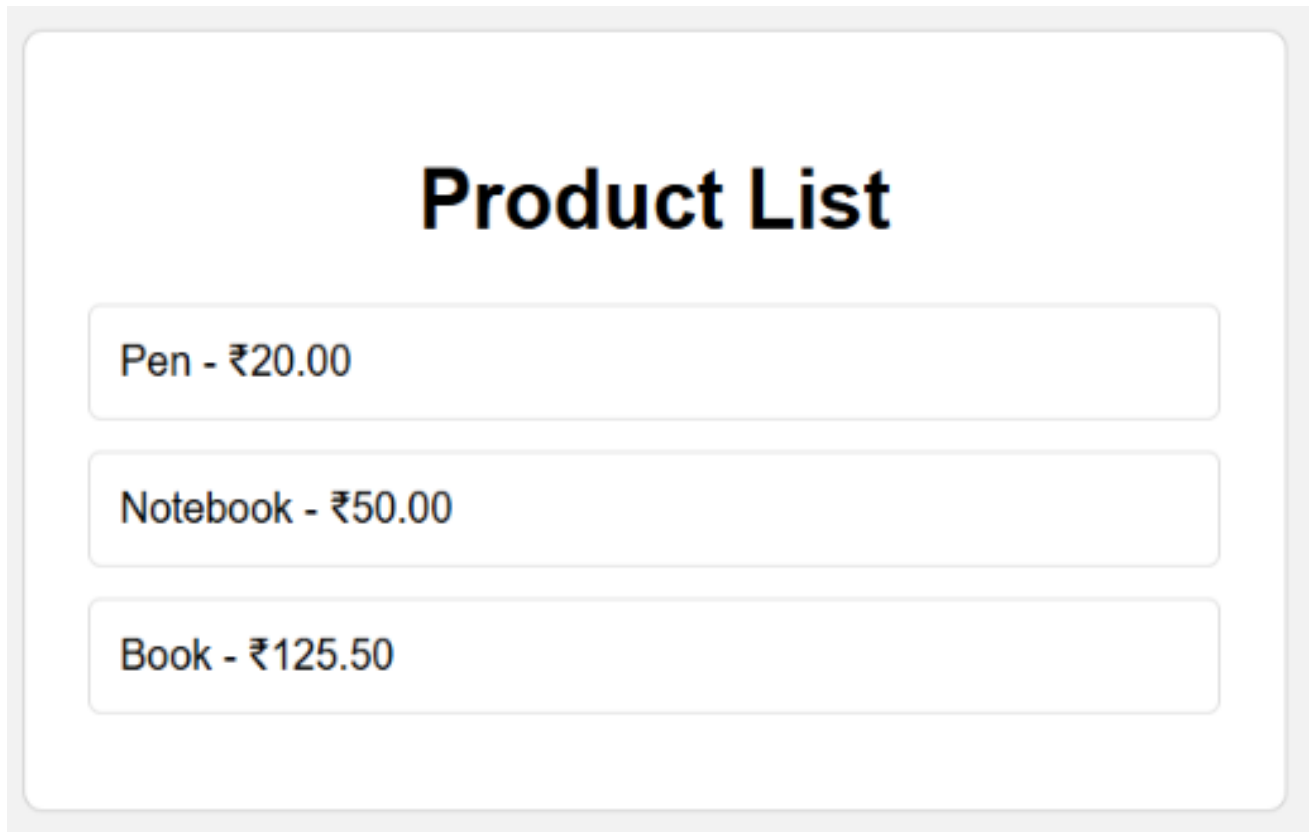
Code	Explanation
textContent	Places the product details into the webpage.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Test	Expected Result	Status
TC-01	Pen, 20	Product List displays Pen - ₹20.00	Pass
TC-02	Notebook, 50	Product List displays Notebook - ₹50.00	Pass
TC-03	Book, 125.50	Product List displays Book - ₹125.50	Pass

9. OUTPUT

Output 1: Program Output Screenshot



10. RESULT

Thus, JavaScript functions and classes were demonstrated by defining a Product class and creating multiple objects with common properties and methods.

How this would be submitted

Source Code

Task_5/

- index.html
- style.css
- script.js
- README.md